Sachi Patel

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Education

Columbia University Expected May 2028
Accelerated M.S. (4+1) in Computer Science, Machine Learning Pathway *New York, NY*

Barnard College of Columbia University Expected May 2027
B.A. in Computer Science & Economics | GPA: 3.83 | New Pathways Bridgewater Scholar *New York, NY*

- **Selected Coursework:** Data Structures, Analysis of Algorithms, Computer Architecture, Systems Programming, Databases, Artificial Intelligence, Applied Machine Learning, Natural Language Processing, Agentic Engineering, Linear Algebra

Skills

Languages/Frameworks: Python, Java, C, TypeScript, JavaScript, SQL, R, MATLAB; React, Next.js, FastAPI, Flask, Streamlit
ML/AI: PyTorch, Google ADK, Hugging Face, LangChain, MCP, RAG, FAISS, scikit-learn, Pandas, NumPy, Ollama
Cloud/Infra: AWS (CDK, ECS Fargate, EC2, S3, Lambda, CloudWatch), Docker, Terraform, GitLab CI/CD
Observability/Data: BigQuery, PostgreSQL, Supabase, OpenTelemetry (OTLP), Prometheus, Grafana, BeautifulSoup

Work Experience

Columbia Build Lab Sep. 2026 – Present
Software Engineer – Kernel *New York, NY*

- Building multi-agent LLM system for entity resolution, reconciling conflicting customer records across pharma data sources
- Developing web-scraping agent that crawls hospital websites and extracts evidence of clinic-to-health-system ownership with LLMs
- Designing source-agnostic ingestion pipeline (Supabase, AWS S3) that normalizes records into a human-in-the-loop review queue

Goldman Sachs Jun. 2026 – Aug. 2026
Software Engineering Intern – Data & AI Solutions Engineering *New York, NY*

- Architected SRE observability stack as infrastructure-as-code (AWS CDK, Terraform) for a data/AI platform serving 15+ teams
- Deployed OTel Collector, Prometheus, Grafana, and CloudWatch Exporter on ECS Fargate to aggregate metrics over OTLP
- Collected 99 application and infrastructure metrics and ingested them into a lakehouse DB to enable long-term reliability analytics
- Built MCP server exposing lakehouse queries to LLM agents, plus agent skills automating metric, dashboard, and alert creation

Depository Trust and Clearing Corporation Jun. 2025 – Aug. 2025
Software Engineering Intern – Cloud Center for Enablement *Jersey City, NJ*

- Built Python automation flagging idle EC2/EBS resources across 25+ SYSIDs; owner alerts drove $350K in annualized savings
- Developed MVP cost-savings dashboard (Python, Flask) pulling CloudAware and AWS Cost Explorer data via REST APIs
- Presented cost findings to CloudOps teams and SYSID owners, driving decommissioning of 200+ idle cloud resources org-wide

Rajiv Sethi Lab, Barnard College Jan. 2024 – Present
Research Assistant *New York, NY*

- Built Python/BeautifulSoup pipeline scraping time-series prices and probabilities from prediction markets across 41 events
- Built backtesting framework (Python, MATLAB) for trading-strategy profitability to compare model vs. market accuracy
- Co-authored paper showing FiveThirtyEight outperformed Polymarket by 17.9% on popular vote; published in ACM CI '25

Barnard Computational Science Center (CSC) Aug. 2024 – Present
Senior Computing Fellow *New York, NY*

- Mentored 120+ students in Python, R, MATLAB across 3 data science courses; led 30+ office hours on debugging and projects
- Designed and taught 3 workshops with modular Colab notebooks covering data wrangling, visualization, and scripting automation

Projects

Agentic Stock Prediction System | *Python, Google ADK, LLMs, Market Data APIs* Jun. 2026 – Present

- Building autonomous trading agent in Google ADK that orchestrates tool calls across market data APIs and technical indicators
- Integrating LLM reasoning over real-time and historical market data to produce buy/sell/hold recommendations with rationale
- Designing evaluation harness that backtests agent recommendations against historical returns to quantify decision quality

RAG-Powered Codebase Assistant | *Python, LangChain, FAISS, Ollama, tree-sitter, Streamlit* Jan. 2026

- Built RAG assistant that ingests GitHub repos, parses Python ASTs via tree-sitter, and chunks code by function and class
- Embedded chunks with sentence-transformers into FAISS; served Ollama LLM answers via LangChain in a Streamlit chat UI
- Scaled indexing across multiple repositories while keeping semantic-search query latency under one second over 10K+ lines of code

HTTP Web Server | *C, POSIX Sockets* Dec. 2024

- Built HTTP/1.0 server in C on POSIX sockets (bind/listen/accept), forking a child process per connection to serve static files
- Implemented request parsing and URI validation with directory-traversal prevention and 200/301/400/404/501 status handling